

- 3 The photograph shows a young locust eating the leaves of a plant.



(Source: © Nigel Cattlin / Alamy Stock Photo)

Locusts can breed in large numbers and destroy crops, such as sorghum, in East Africa.

A scientist observed that some varieties of sorghum were more likely to be eaten by young locusts.

The scientist collected the seeds of two varieties of sorghum (A and B) and grew them in trays.

Locust eggs were collected and hatched into young locusts.

Twenty young locusts were placed in a cage containing 100 g of fresh sorghum leaves of variety A. The cage was kept at 30°C.

After 24 hours, the leaves were removed and the mass of leaves eaten was calculated.

This method was repeated with fresh sorghum leaves of variety B.

The method was repeated six times for these two varieties of sorghum leaves.

The results:

Mass of leaves eaten of variety A

17.3 17.2 17.5 17.0 16.7 16.9

Mass of leaves eaten of variety B

17.8 17.9 17.7 17.6 17.8 17.4

(a) State a suitable null hypothesis for this investigation.

(1)

(b) Draw a suitable table to display the **data** and your calculated **means** for the mass of leaves eaten for these varieties of sorghum.

(3)

- (c) Draw a suitable graph to show the mean mass of leaves eaten of variety A and variety B.

Include an indication of the variability of the data.

(3)



(d) The student analysed the data using the t test formula:

$$t = \frac{(\bar{X}_A - \bar{X}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

Where:

$\bar{x}$ is the mean value for each treatment

n is the number of samples for each treatment

$(S_A)^2 = 0.084$ and $(S_B)^2 = 0.032$

(i) Calculate the value of t .

(2)

Answer

(ii) The table shows the critical values of t for different degrees of freedom.

The number of degrees of freedom = $(n_1 - 1) + (n_2 - 1)$.

Degrees of freedom	p = 0.05	p = 0.01
6	2.45	3.70
7	2.37	3.50
8	2.31	3.36
9	2.26	3.25
10	2.23	3.17
11	2.20	3.11
12	2.18	3.06

Describe the conclusions that can be drawn from this investigation. Use the information in the table to support your answer.

(2)

[illegible]

(e) Describe how the student could extend this investigation to collect more data to either support or reject the hypothesis.

(2)

(f) Some scientists predict that climate change will increase the destruction of crops in East Africa.

Suggest **one** reason why some scientists think this.

(1)

(Total for Question 3 = 14 marks)